

Chapter 40 — Clinical and Urodynamic Evaluation of Lower Urinary Tract Dysfunction in Children

IF YOU READ NOTHING ELSE

1. **Detrusor leak point pressure (DLPP)** >40 cm H₂O → upper-tract risk; <40 safe (McGuire 1981).
2. **Compliance ($\Delta V/\Delta P_{det}$):** >20 normal; 10–20 moderate; <10 severe (mL/cm H₂O).
3. **Expected bladder capacity (EBC):** infant kg × 7; child (age + 2) × 30 mL.
4. **Fill rate** 5–10% EBC/min; stop at Pdet >40 or >150% EBC.
5. **Urodynamics = gold standard;** only if the result changes management.
6. **Electromyography (EMG) lag time** 2–6 s normal; <2 s = detrusor overactivity (DO); >6 s = bladder neck dysfunction.
7. **Post-void residual (PVR) abnormal:** 4–6 y >30 mL / >21%; 7–12 y >20 mL / >15%.
8. **UMPIRE risk:** <25 low; 25–39 or DO intermediate; ≥40 high.
9. **Staccato** never reaches zero; interrupted reaches zero → EMG decides.
10. **Post-urodynamic urinary tract infection (UTI)** 1.4% → no universal culture or prophylaxis.

CLINICAL EVALUATION

- **Secondary enuresis:** new wetting after ≥6 months dry.
- **Sacral roots for bladder and rectum:** S2–S4.
- **Bristol target:** type 3–4, daily and painless; scale 1 (pellets) → 7 (watery).
- **Rome IV constipation:** age ≥4 y, ≥2 of 6 criteria, ≥1 month; ≤2 stools/week, ≥1 soiling/week.
- **ICCS constipation algorithm:** rectum >30 mm → treat constipation; wall >6 mm empty / >3 mm full → treat lower urinary tract symptoms (LUTS).
- **Both improve → maintain** ≥3–6 months.
- **Diaries:** 7-day bowel-bladder diary; 48-hour frequency-volume chart, mandatory if first-line therapy fails.
- **Questionnaires:**
 - **Dysfunctional Voiding Symptom Score (DVSS):** 10 items, max 30; best for following treatment.
 - **Dysfunctional Voiding and Incontinence Scoring System (DVISS):** 14 items, cut-off 9; best for diagnosis.
 - **Paediatric Urinary Incontinence Quality of Life Score (PinQ):** 20 items, max 80; quality of life; does not track response.
 - **Incontinence Symptom Index–Paediatric (ISI-P):** 11 items, max 44, cut-off 9; self-report, age 11–17.
- **Short Screening Instrument for Psychological Problems in Enuresis (SSIPPE):** ≥2 positive → full Child Behavior Checklist (CBCL) → psychology referral.
- **Tri-ponderal mass index = kg/m³,** better than BMI in children; overweight ≥16.0 M / 16.8 F.
- **Neurological level:** high lumbar = hip flexion; mid = knee extension; low = dorsiflexion; sacral = plantar flexion.
- **Absent anal wink or bulbocavernosus reflex** → S2–S4 arc interrupted.
- **Sacral dimple <2.5 cm from anus = low risk** → spinal ultrasound ≤6 months.
- **High-risk stigmata** (vascular lesion, lipoma, hair tuft, multiple, VACTERL) → MRI.
- **Spine ossifies 10–12 months** → ultrasound no longer adequate.
- **Labial fusion:** <2 y; resolves by puberty; topical oestrogen, ~10% recur.
- **Continuous wetting in a girl** → ectopic ureteric orifice.
- **Meatal stenosis:** circumcised boys; treat only if the stream is deviated or narrowed.

LABORATORY TESTING

- **First test in LUTS:** urinalysis; a negative result is reassuring.

- **Culture only if symptomatic** (fever, pain, malaise, new wetting or spasm), especially on clean intermittent catheterisation (CIC).
- **Proteinuria** → repeat; persistent → creatinine ± 24-hour protein.
- **Blood tests** only if upper-tract disease is suspected.

URODYNAMIC EVALUATION

- **ICCS terminology update: 2014.**
- **Uroflow:** voided volume, maximum flow rate (Q_{max}) and pattern matter most; Q_{max} peak ≥2 s.
- **Uroflow valid if toilet-trained, voided volume ≥50% EBC, ≥2 curves.**
- **Patterns:**
 - **tower** = overactive bladder;
 - **staccato** = sphincter over-activity;
 - **interrupted** = underactive detrusor with Valsalva or detrusor-sphincter dyssynergia (DSD);
 - **plateau** = obstruction.
- **Wall thickness:** >3 mm full, >5 mm empty.
- **Bladder volume wall index (BVWI)** normal 70–130%; normal → better desmopressin response.
- **PVR valid only at 50–115% EBC, <5 min after voiding.**
- **Functional capacity** normal ~70–115% EBC.
- **Abnormal capacity (ICCS):** <65% or >150% EBC.
- **Catheter 6 Fr dual-lumen;** EMG patches 3 and 9 o'clock; >1000 Hz sampling.
- **Surface, not needle, EMG.**
- **No general anaesthesia;** ketamine > midazolam if sedation is essential.
- **EBC:** newborn 30–50 mL → young adult 400–500 mL.
- **Stop filling:** urge, pain, voiding, leak > fill, Pdet >40, >150% EBC.
- **Repeat fills:** ICCS 2015 discretionary; first fill worst → report conservatively.
- **Pdet = Pves – Pabd,** zeroed at the pubic symphysis.
- **Abdominal leak point pressure (ALPP), adult female:** <60 intrinsic sphincter deficiency (ISD); 60–90 equivocal; >90 normal.
- **Pressure-flow valid if** ≥50–100 mL voided, ≥50% functional capacity.
- **Pdet at Q_{max}:** boys 66, girls 57; infants up to 118 (boys) / 75 (girls).
- **Danger signs:** poor compliance; DSD with poor emptying; sustained filling Pdet; DLPP >40; high voiding pressure with poor flow.
- **UMPIRE schedule:** baseline 3 months → 1, 2, 3 y → yearly to 5 if vesicoureteric reflux (VUR) or hostile bladder.

- **UMPIRE central vs local agreement ~50%.**

SPECIFIC CLINICAL ENTITIES

- **European Bladder Dysfunction Study (2008):** urodynamics does not predict outcome in non-neurogenic LUTS.
- **Urodynamics if:** neurogenic or obstructive disease; **before reconstruction; refractory** non-neurogenic or cerebral palsy.
- **No urodynamics** as a first-line or routine test in non-neurogenic LUTS.
- **Myelomeningocele = 90% of open dysraphism.**
- **Sacral agenesis:** ~ $\frac{1}{3}$ diagnosed late; flat buttocks; **plain radiograph;** areflexia or DO + DSD.
- **Anorectal malformation → baseline urodynamics ~1 year after repair,** even with a normal cord.
- **Tethered cord:** conus fixed **below L2.**
- **Posterior urethral valves (PUV):** commonest male LUT obstruction; **~15% valve bladder;** DO → poor compliance → myogenic failure.
- **Sacroccygeal teratoma resection → LUT dysfunction up to 50%.**
- **Cerebral palsy:** up to $\frac{1}{3}$; upper motor neuron; **DSD rare.**
- **Spinal cord injury:** shock → retention; **baseline video urodynamics 3–6 months;** higher lesion → more DSD.

- **Transverse myelitis: DO commonest;** motor recovery ≠ bladder recovery; **early CIC.**

CLASSIC TRAPS

- **Trap:** high DLPP and low ALPP mean the same — **Actually:** high DLPP = upper-tract risk; low ALPP = ISD.
- **Trap:** interrupted curve = DSD — **Actually:** it may be Valsalva voiding; EMG decides.
- **Trap:** small bladder with compliance <20 is unsafe — **Actually:** judge by absolute Pdet <30–40.
- **Trap:** a bagged urine culture is reliable — **Actually:** reliable only if negative.
- **Trap:** a late or over-distended PVR is abnormal — **Actually:** it is invalid.
- **Trap:** hydronephrosis resolved, so compliance improved — **Actually:** compliance may not improve.
- **Trap:** DSD was dropped from UMPIRE because it is benign — **Actually:** it was dropped for poor agreement.
- **Trap:** neurological level predicts walking — **Actually:** level and mobility do not correlate.
- **Trap:** high infant voiding pressure = obstruction — **Actually:** up to 118 cm H₂O is physiological in infant boys.
- **Trap:** motor recovery in myelitis means bladder recovery — **Actually:** the bladder usually stays dysfunctional.